

Arithmetic

Chapter F8.5 — Ratio, Percentage, Average, Ages, Mixtures, Time and Work

Module	Foundation Layer — Module 8 — English, Quantitative Aptitude & Reasoning
Chapter	Chapter F8.5
Topic	Official syllabus head 7 — Elementary Mathematics and Statistics
Version	v1.0
Compiled	16 August 2026
Exam stage	Recruitment Test (RT)

HIGH YIELD · WHAT THE 2023 PAPER TOOK FROM THIS CHAPTER

Of the **15 Quantitative Aptitude questions** in 2023, this chapter's topics supplied at least **four**: *ratio and ages, mixtures, and the mean of a set of weights* (PYQ-DERIVED, Blueprint §4.4).

That is more than mensuration, more than algebra, more than data interpretation. **Arithmetic is the largest single slice of the Quant block**, and it is also the cheapest, because every technique in this chapter is school syllabus and none of it requires a formula you have not met before.

Two facts shape how the chapter is written:

1. **The questions are one- or two-step.** This is not a bank-PO paper with three-minute compound arithmetic. F8.0 §1.1 budgets **45 to 60 seconds** for an arithmetic item, and that budget is realistic for everything in §8
2. **Speed comes from the fraction table, not from cleverness.** The single most useful page in Module 8 is **F8.0 §4.1** — the fraction, decimal and percentage equivalents. Almost every percentage question below collapses to a one-step multiplication once you recognise the fraction. §1.2 shows how

Table of Contents

0. How This Chapter Works

1. Session 1 — Percentage ★★★★★

- 1.1 The three questions percentage asks
- 1.2 The fraction method
- 1.3 The four standard shapes
- 1.4 The base — where most marks are lost

2. Session 2 — Ratio and Proportion ★★★★★

- 2.1 The basics, stated precisely
- 2.2 The one-part method
- 2.3 Adding to a ratio — the shape that recurs

3. Session 3 — Averages ★★★★★

- 3.1 The only formula, and its two rearrangements
- 3.2 The replacement rule
- 3.3 Overlapping groups — the subtraction trick

4. Session 4 — Mixtures and Alligation ★★★★★★

- 4.1 The alligation rule
- 4.2 Repeated replacement

5. Session 5 — Problems on Ages ★★★★★★

- 5.1 The method
- 5.2 The two other standard forms

6. Session 6 — Time and Work ★★★★★★

- 6.1 The LCM method — the technique to learn
- 6.2 The rules
- 6.3 Wages — the shape that catches people

7. Session 7 — The Lean Tail ★★★★★★

- 7.1 Average speed
- 7.2 Profit and loss
- 7.3 Simple and compound interest

8. Graded Practice — 35 Questions ★★★★★

- 8.1 Level 1 — Single step
- 8.2 Level 2 — Two steps
- 8.3 Level 3 — Recruitment Test standard

9. Answers and Explanations

9.1 Level 1

9.2 Level 2

9.3 Level 3

10. One-Minute Revision

11. Five-Minute Wall Sheet

0. How This Chapter Works

STUDY SESSION · CHAPTER F8.5

1. **Session 1** — Percentage, and the fraction method (§1) ★★★★★
2. **Session 2** — Ratio and proportion (§2) ★★★★★
3. **Session 3** — Averages (§3) ★★★★★
4. **Session 4** — Mixtures and alligation (§4) ★★★★★★
5. **Session 5** — Problems on ages (§5) ★★★★★★
6. **Session 6** — Time and work (§6) ★★★★★★
7. **Session 7** — The lean tail: speed, profit, interest (§7) ★★★★★★

Then **35 graded questions** in §8 — the largest bank in Module 8, because this is the largest slice of the Quant block.

F8.0's calendar takes §8 Q1–12 on Tuesday of week 1, Q13–24 on Thursday of week 1, and Q25–35 on Tuesday of week 4, then works the errors in weeks 6 and 8, and takes all 35 timed in week 10.

MUST MASTER · THE THREE HABITS THAT DECIDE THIS CHAPTER

1. **Convert every percentage to a fraction before you multiply.** $16\frac{2}{3}\%$ is $\frac{1}{6}$, $12\frac{1}{2}\%$ is $\frac{1}{8}$, $37\frac{1}{2}\%$ is $\frac{3}{8}$. A question asking for a $16\frac{2}{3}\%$ increase is asking you to multiply by **$\frac{7}{6}$** — one step, no decimals
2. **Give the ratio a unit.** Never carry 4:3 through a problem as two numbers. Write **4x and 3x**, or better, note that one part is worth a definite amount and find that amount first. §2.2
3. **Work in units of work, not fractions of a day.** For time and work, take the **LCM of the given days as the total work**. A job done by A in 12 days and B in 18 becomes 36 units of work at 3 and 2 units a day. No fractions appear anywhere. §6.1

Habit 3 alone will halve your time on §6, and it is the technique most candidates never learn.

1. Session 1 — Percentage ★★★★★

1.1 The three questions percentage asks

Question	Method
What is x% of N?	Convert x% to a fraction, multiply
A is what per cent of B?	$(A/B) \times 100$
A is what per cent more or less than B?	$(\text{difference} \div B) \times 100$ — the base is what you compare to

1.2 The fraction method

The percentages that appear in examinations are not random. They are the ones with clean fractional equivalents, because the examiner also wants the arithmetic to close.

Percentage	Fraction	Percentage	Fraction
50%	$1/2$	$12\frac{1}{2}\%$	$1/8$
$33\frac{1}{3}\%$	$1/3$	$11\frac{1}{9}\%$	$1/9$
25%	$1/4$	$9\frac{1}{11}\%$	$1/11$
20%	$1/5$	$8\frac{1}{3}\%$	$1/12$
$16\frac{2}{3}\%$	$1/6$	$6\frac{1}{4}\%$	$1/16$
$14\frac{2}{7}\%$	$1/7$	5%	$1/20$
$37\frac{1}{2}\%$	$3/8$	$62\frac{1}{2}\%$	$5/8$
$66\frac{2}{3}\%$	$2/3$	$83\frac{1}{3}\%$	$5/6$

Worked. A basic wage of ₹18,000 rises by $16\frac{2}{3}\%$. New wage = $18,000 \times \frac{7}{6} = \text{₹}21,000$. Compare the decimal route: $18,000 \times 0.1667 = 3,000.06$, then add. The fraction route is faster and exact.

1.3 The four standard shapes

MUST MASTER · THE FOUR PERCENTAGE RESULTS WORTH KNOWING BY HEART

1. A is $x\%$ more than B, so B is less than A by $x/(100 + x) \times 100$. If A is 25% more than B, then B is $25/125 = 20\%$ less than A. The two percentages are **never the same**, because the base changes. This is the commonest percentage trap in existence — Q1

2. Increase by $x\%$ then decrease by $x\%$ is always a net LOSS of $x^2/100$ per cent. Up 20% then down 20% leaves $0.8 \times 1.2 = 0.96$, a **4% fall** — never "no change" — Q8

3. If the price rises by $x\%$, consumption must fall by $x/(100 + x) \times 100$ to keep expenditure the same. A 25% rise needs a **20% cut**, not 25% — Q28

4. Two successive discounts of $a\%$ and $b\%$ equal a single discount of $(a + b - ab/100)$ per cent. 20% and 10% together give 28%, not 30% — Q13

All four are the same underlying point: **percentages do not add, because the base moves.** Multiply the factors instead — $0.8 \times 0.9 = 0.72$ — and read the answer off the product.

1.4 The base — where most marks are lost

A is 20% of B and *B is 20% more than A* involve completely different bases. Before computing, name the base in one word: **per cent of what?** If the question says "*increased by*", the base is the original value; if it says "*of*", the base is the noun that follows.

CHECK YOURSELF · SESSION 1

1. Give fractions for $16\frac{2}{3}\%$, $12\frac{1}{2}\%$, $37\frac{1}{2}\%$, $66\frac{2}{3}\%$, $8\frac{1}{3}\%$
2. If A is 25% more than B, by what per cent is B less than A?
3. A sum rises 10% then falls 10%. Net change?
4. Price rises 20%; by how much must consumption fall to hold expenditure constant?
5. Single equivalent of successive discounts of 25% and 20%?

2. Session 2 — Ratio and Proportion ★★★★★

2.1 The basics, stated precisely

A ratio compares quantities **of the same kind** and has no units. $a : b$ is the fraction a/b , and it is unchanged if both terms are multiplied by the same number — which is why every ratio problem is solved by finding the value of **one part**.

Idea	Statement
Proportion	$a : b = c : d$, so $ad = bc$ — the product of the extremes equals that of the means
Compounded ratio	Combine $a:b$ and $b:c$ by making the common term equal: $a:b:c$
Duplicate / sub-duplicate	$a^2:b^2$ · and the square-root ratio
Dividing a sum	In $a : b : c$, each share is <i>its part</i> ÷ <i>total parts</i> of the sum
Direct proportion	More men, more work: multiply
Inverse proportion	More men, fewer days : divide

2.2 The one-part method

MUST MASTER · FIND WHAT ONE PART IS WORTH, THEN ANSWER

Almost every ratio question in this examination reduces to a single line of arithmetic once you find the value of one part.

Worked — a sum divided. ₹6,300 is divided in the ratio 2 : 3 : 4. Total parts = 9, so **one part = 700**, and the shares are 1,400, 2,100 and 2,800.

Worked — a difference given. A and B share a sum in the ratio 5 : 3, and A gets ₹800 more than B. The **difference is 2 parts**, so one part = 400, and the total, being 8 parts, is **₹3,200**. Note that the question never told you the total — the difference did.

Worked — combining ratios. $a : b = 2 : 3$ and $b : c = 4 : 5$. Make b common: multiply the first by 4 and the second by 3 to get $a : b : c = 8 : 12 : 15$, so $a : c = 8 : 15$.

The discipline: write down what one part is worth **before** you look at the options. Candidates who skip that step answer with a part instead of a total, which is exactly the wrong option offered.

2.3 Adding to a ratio — the shape that recurs

When a quantity is **added to one component only**, the other component is unchanged, and that unchanged quantity is your anchor.

Worked. 45 litres of a mixture is milk and water in the ratio 4 : 1, so milk = 36 L and water = 9 L. How much water makes it 3 : 2? **The milk stays 36.** In a 3 : 2 mixture with 36 of milk, water must be

1. So **add 15 litres.** Q31 is this question.

TRAP · THE UNCHANGED QUANTITY IS THE ONE TO FIX

When water is added, milk is constant. When one salary is raised, the other is constant. Set up the new ratio **in terms of the constant**, not in terms of a new variable.

Candidates who write "let x litres be added" and solve $(36)/(9 + x) = 3/2$ arrive at the same answer, but slower and with more chances to slip. Anchoring on the constant makes it mental arithmetic.

CHECK YOURSELF · SESSION 2

1. Divide ₹8,400 in the ratio 3 : 4 : 5
2. $A : B = 3 : 4$ and $B : C = 6 : 7$. Find $A : C$
3. Two shares are in the ratio 7 : 4 and differ by ₹900. What is the total?
4. In 60 litres of a 5 : 1 milk–water mixture, how much water makes it 5 : 2?

3. Session 3 — Averages ★★★★★

3.1 The only formula, and its two rearrangements

Average = total ÷ number of items. Therefore **total = average × number**, and that second form is the one you actually use.

Almost every average question is solved by converting averages into totals, doing the arithmetic on totals, and converting back. The word "average" in a question is an instruction to multiply.

Shape	Method
A member is replaced	Change in total = number × change in average
A member is added	New total - old total = the new member's value
Overlapping groups	Add the group totals and subtract the whole, to isolate the overlap
A wrong reading is corrected	Change in average = (error ÷ number of items)

3.2 The replacement rule

MUST MASTER · CHANGE IN TOTAL = NUMBER × CHANGE IN AVERAGE

This one line answers the 2023 "*mean of weights*" family in a single step.

Worked. The average weight of 8 men is 62 kg. A man weighing 55 kg is replaced and the average becomes 63 kg. The average rose by 1 kg over 8 men, so the **total rose by 8 kg**, so the new man weighs $55 + 8 = \mathbf{63 \text{ kg}}$.

No equations, no unknowns. The same rule handles a member being **added**: if 30 members average 32 years and adding one raises the average to 33, the newcomer's age is $31 \times 33 - 30 \times 32 = \mathbf{63}$ — Q16.

And corrections: if one observation of 96 was read as 69, the total is short by 27, so over 25 observations the average is short by $27/25 = \mathbf{1.08}$ — Q30.

3.3 Overlapping groups — the subtraction trick

Worked. A, B and C average 45 kg. A and B average 40; B and C average 43. Find B.

$(A + B) + (B + C) = 80 + 86 = 166$, and $A + B + C = 135$. Subtracting, **B = 31 kg**.

The pattern generalises: where two overlapping pairs and the whole are given, **the sum of the pairs minus the whole leaves the overlap**. Q25 and Q10 are both this shape.

CHECK YOURSELF · SESSION 3

1. State the replacement rule
2. Average of 10 numbers is 45; one number 27 is replaced by 47. New average?
3. The average of 11 results is 50, of the first six 49, of the last six 52. Find the sixth
4. Average of 20 observations was 45; an observation of 61 was read as 16. Correct average?

4. Session 4 — Mixtures and Alligation ★★★★★

4.1 The alligation rule

For **two** ingredients mixed to a known mean value:

$$(\text{quantity of cheaper}) : (\text{quantity of dearer}) = (\text{price of dearer} - \text{mean}) : (\text{mean} - \text{price of cheaper})$$

The differences are taken **crosswise**, and the result is a ratio of **quantities**.

Worked. Milk at ₹40 a litre is mixed with water, which is free, so that the mixture costs ₹32 a litre. Ratio of milk to water = $(32 - 0) : (40 - 32) = 32 : 8 = \mathbf{4 : 1}$.

Worked, with two mixtures. Vessel I has milk and water in the ratio 5 : 2, vessel II in the ratio 3 : 4. In what ratio must they be mixed to get equal milk and water? Work in **milk fractions**: $5/7$ and $3/7$, target $1/2$. Ratio = $(1/2 - 3/7) : (5/7 - 1/2) = 1/14 : 3/14 = \mathbf{1 : 3}$.

MUST MASTER · ALWAYS ALLIGATE ON THE SAME COMPONENT

The single mechanical error in alligation is mixing the components: taking the milk fraction of one vessel and the water fraction of the other.

Choose one component — milk, or alcohol, or the cheaper item — and express all three values as that component's share. Then apply the crosswise rule.

And note what the answer is: a ratio of the **quantities of the two mixtures**, not of their components. Q27 offers the reversed ratio 3 : 1 as an option for exactly this reason.

4.2 Repeated replacement

When y units are drawn from x units of pure liquid and replaced by water, and this is done n times:

$$\text{Liquid remaining} = x \times [(x - y)/x]^n$$

Worked. 20 litres of milk; 4 litres drawn off and replaced by water, twice. Milk left = $20 \times (16/20)^2 = 20 \times 0.64 = \mathbf{12.8 \text{ litres}}$ — Q19.

Note the trap: the formula gives the **liquid remaining**, not the amount removed, and it applies only when the volume is restored each time.

CHECK YOURSELF · SESSION 4

1. State the alligation rule
2. In what ratio must rice at ₹30 and ₹45 a kg be mixed to sell at ₹35?
3. From 40 litres of milk, 8 litres are removed and replaced by water, twice. Milk left?
4. Why must alligation be done on one component only?

5. Session 5 — Problems on Ages ★★★★★

5.1 The method

Ages are ratio problems with a time shift. Three rules make them mechanical:

1. **Let the present ages be the ratio with an x** — $7x$ and $3x$, not two separate unknowns. One variable, one equation
2. **Add or subtract the shift from EVERY age.** Ten years hence, both are ten years older
3. **Form the equation from the second ratio** and cross-multiply

Worked — the 2023 ratio-and-ages shape. The ages of a father and son are in the ratio $7 : 3$, and ten years hence the ratio will be $2 : 1$. Then $(7x + 10)/(3x + 10) = 2/1$, so $7x + 10 = 6x + 20$, giving $x = 10$ and the father's present age **70**. Check: 70 and 30 become 80 and 40, which is $2 : 1$ — Q26.

MUST MASTER · CHECK BY SUBSTITUTION, IT COSTS FIVE SECONDS

Age problems are the easiest place in the Quant paper to make a sign error, and the **only** place where verification is completely free: put your answer back into the words of the question.

If the father is 70 and the son 30, then in ten years they are 80 and 40 — a ratio of $2 : 1$, as required. Five seconds, and it converts a probable mark into a certain one.

The examiner's standard trap is to offer the value of x , or the son's age, when the father's was asked. In the worked example, $x = 10$ and the son is 30; both are wrong answers to the question asked. Before you mark, re-read **whose** age was wanted.

5.2 The two other standard forms

Form	Example	Equation
A multiple now, another later	A father is 3 times his son's age; in 12 years, twice	$3s + 12 = 2(s + 12)$, so $s = 12$
A multiple now, another earlier	3 times now; 15 years ago, 9 times	$3s - 15 = 9(s - 15)$, so $s = 20$

Both are one linear equation. Note in the second that the larger multiple belongs to the **earlier** time — ratios of ages always move **towards $1 : 1$** as people get older, which is a useful sanity check on any answer.

CHECK YOURSELF · SESSION 5

1. Ages are 5 : 2; in 6 years, 2 : 1. Find both
2. A man is 4 times his son; in 20 years, twice. Find both
3. Why must a ratio of ages move towards 1 : 1 over time?
4. What are the two wrong answers an age question usually offers?

6. Session 6 — Time and Work ★★★★★

6.1 The LCM method — the technique to learn

MUST MASTER · TAKE THE TOTAL WORK AS THE LCM OF THE DAYS

Do not add fractions. Take the **total work as the LCM of the given numbers of days**, and every rate becomes a whole number of units per day.

Worked. A does a job in 12 days, B in 18. LCM = **36 units** of work. A does 3 units a day, B does 2. Together 5 a day, so they finish in $36/5 = 7.2$ days — Q7.

Worked, one leaves. A in 20 days, B in 30. LCM = **60 units**; A does 3 a day, B does 2, together 5. In 6 days they complete 30 units — exactly half. The remaining 30 units at A's 3 a day take **10 days** — Q29.

Worked, efficiency given. A is twice as efficient as B and together they take 12 days. Let B do 1 unit a day and A 2, so together 3 a day and the job is 36 units. **B alone: 36 days** — Q21.

Every fraction has disappeared. This is the single highest-return technique in the chapter, and it also prevents the commonest error in the topic — **adding days instead of adding rates**.

6.2 The rules

Situation	Rule
Together	Add the rates , never the days
A alone, from together	B's rate = (together's rate) - (A's rate)
Efficiency	Twice as efficient means half the time ; rates are in the ratio of efficiencies
Wages	Wages divide in the ratio of work done , not of days worked
Men and days	$M_1D_1/W_1 = M_2D_2/W_2$ — more men, fewer days
Pipes	An emptying pipe is a negative rate

6.3 Wages — the shape that catches people

Worked. A and B contract a job for ₹2,400. A alone would take 20 days, B 30. With C's help they finish in 10 days. What is C's share?

Take 60 units. A does 3 a day, B 2, so $A + B = 5$. To finish in 10 days the team must do 6 a day, so **C does 1 unit a day**. Over 10 days: A 30 units, B 20, **C 10** — a sixth of the work, so **₹400** — Q35.

The trap is to divide the money by three, or in the ratio of days. **Wages follow work done.**

CHECK YOURSELF · SESSION 6

1. Why take the LCM as the total work?
2. A in 15 days, B in 10. How long together?
3. A and B together in 8 days, A alone in 12. B alone?
4. In what ratio do wages divide?

7. Session 7 — The Lean Tail ★★☆☆☆

Three topics that are **not** in the 2023 evidence but are one step from what is. They get one session between them, and that is a deliberate judgement about yield, not an oversight: percentage is the parent of all three, so if you have \$1 you have most of this already.

7.1 Average speed

Average speed = total distance ÷ total time. Never the average of the speeds.

For **equal distances** at speeds u and v , average speed = $2uv/(u + v)$ — the harmonic mean.

Worked. Half the distance at 40 km/h, half at 60. Average = $2 \times 40 \times 60/100 = 48$ km/h, not 50 — Q23. It is below 50 because more time is spent at the slower speed.

For **equal times**, the average is the arithmetic mean. **Equal distances give the harmonic mean; equal times give the arithmetic mean** — and the question always tells you which by saying *half the distance* or *half the time*. F9.1 §2.2's conversion applies here: **km/h to m/s is $\times 5/18$** .

7.2 Profit and loss

Quantity	Formula
Profit %	$(SP - CP)/CP \times 100$ — always on cost
SP from a gain	$SP = CP \times (100 + \text{gain\%})/100$
CP from SP	$CP = SP \times 100/(100 + \text{gain\%})$
Discount	On the marked price , not on cost

TRAP · EQUAL SELLING PRICES, EQUAL PERCENTAGES, ONE GAIN ONE LOSS » ALWAYS A LOSS

Two articles sold at the **same price**, one at a gain of $x\%$ and one at a loss of $x\%$, always produce an **overall loss of $x^2/100$ per cent**.

Worked. Two articles at ₹1,200 each, one at +20%, one at -20%. Cost prices are $1,200 \times 100/120 = 1,000$ and $1,200 \times 100/80 = 1,500$, so total cost 2,500 against total sale 2,400: a loss of ₹100 on ₹2,500 = **4%** — which is $20^2/100$ — Q33.

The reason is the same as §1.3's second result: **the two percentages have different bases**. The loss is taken on the larger cost price, so it outweighs the gain. "No profit no loss" is the wrong option waiting for you.

7.3 Simple and compound interest

SI = $PRT/100$. CI: amount = $P(1 + R/100)^t$, and CI = amount - P.

Result	Value
Difference between CI and SI, 2 years	$P(R/100)^2$
Difference, 3 years	$P(R/100)^2 \times (300 + R)/100$
Compounded half-yearly	Rate halves, time doubles
Money doubles under SI	When $RT = 100$

Worked. ₹10,000 for 2 years at 10%. SI = 2,000. CI = $10,000 \times 1.21 - 10,000 = 2,100$. Difference **₹100**, which is $10,000 \times (10/100)^2 = Q32$.

The two-year difference is worth memorising, because it converts a two-line computation into a one-step multiplication, and it is the only form of the question that is commonly set.

CHECK YOURSELF · SESSION 7

1. Average speed for equal distances at 30 and 60 km/h?
2. Why is profit percentage always taken on cost?
3. Two articles sold at the same price, +25% and -25%. Overall?
4. Difference between CI and SI on ₹20,000 for 2 years at 5%?

8. Graded Practice — 35 Questions ★★★★★

Target: 45 to 60 seconds each. Convert percentages to fractions, take the LCM for work problems, and name the base before you divide.

8.1 Level 1 — Single step

Q1. If A's salary is 25% more than B's, then B's salary is less than A's by (a) 25% (b) $22\frac{1}{2}\%$ (c) 20% (d) $16\frac{2}{3}\%$

Q2. An increase of $16\frac{2}{3}\%$ is the same as multiplying by (a) $\frac{7}{6}$ (b) $\frac{6}{7}$ (c) $\frac{5}{6}$ (d) $\frac{8}{7}$

Q3. ₹6,300 is divided among three persons in the ratio 2 : 3 : 4. The largest share is (a) ₹1,400 (b) ₹2,100 (c) ₹2,520 (d) ₹2,800

Q4. The average weight of 8 men is 62 kg. One man weighing 55 kg is replaced by another and the average becomes 63 kg. The weight of the new man is (a) 62 kg (b) 63 kg (c) 70 kg (d) 71 kg

Q5. The ages of A and B are in the ratio 4 : 3 and their sum is 42 years. A's age is (a) 18 years (b) 21 years (c) 24 years (d) 28 years

Q6. 40 litres of a mixture contain milk and water in the ratio 3 : 1. The quantity of water is (a) 10 litres (b) 12 litres (c) 15 litres (d) 30 litres

Q7. A can do a piece of work in 12 days and B in 18 days. Working together they will finish it in (a) 6 days (b) 7.2 days (c) 7.5 days (d) 15 days

Q8. A sum is first reduced by 20% and then increased by 20%. The net change is (a) no change (b) an increase of 4% (c) a decrease of 2% (d) a decrease of 4%

Q9. If $a : b = 2 : 3$ and $b : c = 4 : 5$, then $a : c$ is (a) 8 : 15 (b) 2 : 5 (c) 3 : 4 (d) 8 : 12

Q10. The average of 11 results is 50. The average of the first six is 49 and of the last six is 52. The sixth result is (a) 48 (b) 50 (c) 56 (d) 61

Q11. A father is three times as old as his son. After 12 years he will be twice as old as his son. The son's present age is (a) 10 years (b) 12 years (c) 14 years (d) 16 years

Q12. An employee's share of contribution is 12% of basic wages of ₹18,000. The amount deducted is (a) ₹1,800 (b) ₹1,980 (c) ₹2,100 (d) ₹2,160

8.2 Level 2 — Two steps

Q13. Two successive discounts of 20% and 10% are equivalent to a single discount of (a) 28% (b) 30% (c) 26% (d) 25%

Q14. In what ratio must milk costing ₹40 a litre be mixed with water, which costs nothing, so that the mixture costs ₹32 a litre? (a) 3 : 1 (b) 5 : 1 (c) 4 : 1 (d) 4 : 5

Q15. A and B together can complete a work in 8 days, and A alone can complete it in 12 days. B alone will take (a) 20 days (b) 24 days (c) 16 days (d) 32 days

- Q16.** The average age of 30 members of a committee is 32 years. When the secretary is included the average rises by one year. The secretary's age is (a) 33 years (b) 45 years (c) 52 years (d) 63 years
- Q17.** A sum is divided between A and B in the ratio 5 : 3, and A receives ₹800 more than B. The total sum is (a) ₹3,200 (b) ₹2,400 (c) ₹4,000 (d) ₹1,600
- Q18.** A candidate scoring 35% of the total marks fails by 40 marks, while one scoring 45% obtains 20 marks more than the pass mark. The total marks are (a) 500 (b) 550 (c) 600 (d) 650
- Q19.** A vessel contains 20 litres of milk. Four litres are drawn off and replaced by water, and the same is done once again. The milk now remaining is (a) 12 litres (b) 12.8 litres (c) 13.2 litres (d) 14.4 litres
- Q20.** The present ages of A and B are in the ratio 3 : 5. Five years ago the ratio was 1 : 2. A's present age is (a) 15 years (b) 18 years (c) 20 years (d) 25 years
- Q21.** A is twice as efficient as B, and working together they complete a job in 12 days. B alone would take (a) 18 days (b) 24 days (c) 30 days (d) 36 days
- Q22.** Contribution is payable at 12% of basic wages. If basic wages rise by 25%, the contribution rises by (a) 12% (b) 15% (c) 25% (d) 37%
- Q23.** A man covers half of a journey at 40 km/h and the other half at 60 km/h. His average speed for the whole journey is (a) 50 km/h (b) 48 km/h (c) 45 km/h (d) 52 km/h
- Q24.** If 12 men can complete a work in 15 days, the number of days 20 men will take is (a) 9 (b) 10 (c) 12 (d) 25

8.3 Level 3 — Recruitment Test standard

- Q25.** The average weight of A, B and C is 45 kg. The average weight of A and B is 40 kg and that of B and C is 43 kg. The weight of B is (a) 17 kg (b) 20 kg (c) 26 kg (d) 31 kg
- Q26.** The ages of a father and his son are in the ratio 7 : 3. Ten years hence the ratio will be 2 : 1. The father's present age is (a) 30 years (b) 60 years (c) 70 years (d) 84 years
- Q27.** Two vessels contain milk and water in the ratios 5 : 2 and 3 : 4 respectively. In what ratio must their contents be mixed so that the resulting mixture has milk and water in equal measure? (a) 3 : 1 (b) 1 : 3 (c) 2 : 3 (d) 7 : 5
- Q28.** The price of a commodity rises by 25%. The percentage by which consumption must be reduced so that expenditure remains unchanged is (a) 20% (b) 25% (c) $16\frac{2}{3}\%$ (d) $33\frac{1}{3}\%$
- Q29.** A can do a piece of work in 20 days and B in 30 days. They work together for 6 days, after which B leaves. A will complete the remaining work in a further (a) 6 days (b) 8 days (c) 10 days (d) 12 days
- Q30.** The average of 25 observations was found to be 78. It was later discovered that an observation of 96 had been recorded as 69. The correct average is (a) 78.92 (b) 79.00 (c) 79.04 (d) 79.08
- Q31.** A 45-litre mixture contains milk and water in the ratio 4 : 1. The quantity of water that must be added to make the ratio 3 : 2 is (a) 9 litres (b) 15 litres (c) 18 litres (d) 24 litres
- Q32.** The difference between the compound interest and the simple interest on ₹10,000 for two years at 10% per annum is (a) ₹100 (b) ₹110 (c) ₹121 (d) ₹200

- Q33.** A dealer sells two articles at ₹1,200 each, gaining 20% on one and losing 20% on the other. On the transaction as a whole he makes (a) neither profit nor loss (b) a profit of 4% (c) a loss of 4% (d) a loss of ₹200
- Q34.** A man is at present three times as old as his son. Fifteen years ago he was nine times as old. The man's present age is (a) 45 years (b) 60 years (c) 54 years (d) 72 years
- Q35.** A and B undertake a piece of work for ₹2,400. A alone could do it in 20 days and B alone in 30 days. With the help of C they complete it in 10 days. C's share of the money is (a) ₹300 (b) ₹360 (c) ₹380 (d) ₹400

9. Answers and Explanations

9.1 Level 1

Q	Ans	Explanation
1	c	$A = 1.25B$, so the difference is $0.25B$ on a base of A , not B: $0.25/1.25 = 20\%$. §1.3 result 1. Option (a) is the answer to a question about the other direction
2	a	$16\frac{2}{3}\% = \frac{1}{6}$, so an increase multiplies by $1 + \frac{1}{6} = \frac{7}{6}$. This is the fraction habit that pays throughout the chapter
3	d	$2 + 3 + 4 = 9$ parts; one part = $6,300/9 = 700$; largest = $4 \times 700 = ₹2,800$. Note (b) is 3 parts and (a) is 2 — the wrong-share trap
4	b	The average rose 1 kg over 8 men, so the total rose 8 kg : $55 + 8 = 63$ kg. §3.2's rule in one line, and the 2023 "mean of weights" shape
5	c	$4 + 3 = 7$ parts for 42 years, so one part = 6 and $A = 4 \times 6 = 24$. (a) is B's age — always check whose was asked
6	a	Water is 1 part of 4: $40 \times \frac{1}{4} = 10$ litres. (d) is $\frac{3}{4}$, the milk
7	b	LCM 36 units; A does 3 a day, B 2, together 5. $36/5 = 7.2$ days. Adding the days to get 30, or averaging them to 15, are the two standard errors — (d) is the average
8	d	$0.8 \times 1.2 = 0.96$, a 4% decrease . §1.3 result 2: down-then-up by the same percentage is always a loss of $x^2/100$, here $400/100$. (a) is the intuitive wrong answer
9	a	Make b common: $a : b = 8 : 12$ and $b : c = 12 : 15$, so $a : b : c = 8 : 12 : 15$ and $a : c = 8 : 15$
10	c	$6 \times 49 + 6 \times 52 = 606$, against a whole of $11 \times 50 = 550$. The sixth result is counted twice , so it is $606 - 550 = 56$. §3.3's overlap trick
11	b	$3s + 12 = 2(s + 12)$ gives $s = 12$. Check: 36 and 12 become 48 and 24, which is $2 : 1$
12	d	12% of 18,000 = $18,000 \times \frac{3}{25} = ₹2,160$

9.2 Level 2

Q	Ans	Explanation
13	a	$0.8 \times 0.9 = 0.72$, so a single discount of 28% . (b) is the sum of the two discounts, which is always wrong because the second discount applies to a reduced price
14	c	Alligation on price: $(32 - 0) : (40 - 32) = 32 : 8 = 4 : 1$. Sense check: the mixture costs 4/5 of the milk price, so it must be 4/5 milk
15	b	LCM 24 units; together 3 a day, A 2 a day, so B does 1 a day and takes 24 days . Subtracting days (8 from 12) to get 4 is the classic error
16	d	$31 \times 33 - 30 \times 32 = 1,023 - 960 = 63$ years . (a) is the new average, offered for candidates who stop one step early
17	a	The difference is $5 - 3 = 2$ parts = ₹800, so one part = 400 and the total of 8 parts = ₹3,200 . S2.2's second worked case — the total was never stated
18	c	Both expressions equal the pass mark: $0.35T + 40 = 0.45T - 20$, so $0.10T = 60$ and $T = 600$. The pass mark, incidentally, is 250
19	b	$20 \times (16/20)^2 = 20 \times 0.64 = 12.8$ litres . (a) is what you get by subtracting 4 twice and forgetting that the second draw removes diluted milk, not pure milk
20	a	$(3x - 5)/(5x - 5) = 1/2$ gives $6x - 10 = 5x - 5$, so $x = 5$ and $A = 15$. Check: 15 and 25 were 10 and 20, which is 1 : 2
21	d	Let B do 1 unit a day and A 2, so 3 a day for 12 days = 36 units. B alone: 36 days . (a) is A's time, (b) is the trap of halving
22	c	The contribution is a fixed proportion of wages, so a 25% rise in wages is a 25% rise in the contribution. The 12% is a distractor and never enters the arithmetic — a conceptual item, not a computational one
23	b	Equal distances , so the harmonic mean: $2 \times 40 \times 60/100 = 48$ km/h . (a) is the arithmetic mean, correct only for equal times
24	a	Inverse proportion: $12 \times 15 = 20 \times D$, so $D = 9$ days

9.3 Level 3

Q	Ans	Explanation
25	d	$(A + B) + (B + C) = 80 + 86 = 166$ and $A + B + C = 135$, so B = 31 kg . The 2023 mean-of-weights item in its harder form; §3.3
26	c	$(7x + 10)/(3x + 10) = 2$ gives $x = 10$, so the father is 70 . Check: 80 and 40. Note (a) is the son's present age and (b) is $6x$ — both are offered, per §5.1
27	b	Alligate on milk : $5/7$ and $3/7$ against a target of $1/2$. $(1/2 - 3/7) : (5/7 - 1/2) = 1/14 : 3/14 = 1 : 3$. (a) is the same numbers reversed, which is what you get by alligating the wrong way round
28	a	$25/(100 + 25) = 1/5 = 20\%$. §1.3 result 3 — a 25% rise needs a 20% cut, because expenditure is price $\times$ quantity and the base has moved. (b) is the intuitive error
29	c	LCM 60; A 3 a day, B 2, together 5. Six days completes 30 units, half the work. The remaining 30 at A's 3 a day takes 10 days
30	d	The total was short by $96 - 69 = 27$, so the average was short by $27/25 = 1.08$, giving 79.08
31	b	Milk = 36 L and is unchanged . For a 3 : 2 ratio with 36 of milk, water must be 24, so add 15 litres . §2.3 — anchor on the constant. (d) is the new total water, not the addition
32	a	$P(R/100)^2 = 10,000 \times 1/100 = ₹100$. Longhand: CI 2,100 against SI 2,000. (c) is $10,000 \times (1.1^2 - 1)$ written carelessly
33	c	Cost prices are 1,000 and 1,500, total 2,500 against sales of 2,400: a loss of ₹100 on 2,500 = 4% , which is $20^2/100$. §7.2 — (a) is the intuitive answer and (d) doubles the actual rupee loss
34	b	$3s - 15 = 9(s - 15)$ gives $6s = 120$, $s = 20$, so the man is 60 . Check: 45 and 5, which is 9 times
35	d	60 units; A 3 a day, B 2. To finish in 10 days the team must do 6 a day, so C does 1 — 10 of the 60 units, a sixth. $2,400/6 = ₹400$. Wages follow work done , so dividing by three gives 800, which is not offered, and dividing by days gives nonsense

THINK LIKE UPSC · THE FIVE SHAPES IN THIS BANK

1. **The moving base.** Q1, Q8, Q13, Q28, Q33. **Five of thirty-five questions** are the same insight — percentages cannot be added or reversed, because the base changes. Multiply the factors instead. Learning this one idea properly is worth more than any other page in the chapter
2. **The wrong quantity, correctly computed.** Q3 offers 2 parts and 3 parts; Q5 offers B's age; Q16 offers the new average; Q26 offers the son's age; Q31 offers the new total water. In every case the arithmetic is right and the **question was different**. Before marking, re-read what was asked
3. **Averages as totals.** Q4, Q10, Q16, Q25, Q30. Never work with averages directly — multiply up to totals, do the arithmetic, divide back
4. **LCM instead of fractions.** Q7, Q15, Q21, Q29, Q35. Five questions, no fractions needed in any of them
5. **The intuitive answer offered first.** Q8(a) "no change", Q13(b) "30%", Q23(a) "50", Q28(b) "25%", Q33(a) "neither profit nor loss". In each case option (a) or (b) is what a candidate answers **without calculating**. Where an option looks like the obvious symmetry, that is the reason to compute

10. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F8.5

WEIGHT the largest slice of the 15-question Quant block — **ratio and ages, mixtures, mean of weights** all appeared in 2023 · **target 45–60 seconds**

FRACTIONS $16\frac{2}{3}\% = 1/6$ · $12\frac{1}{2}\% = 1/8$ · $37\frac{1}{2}\% = 3/8$ · $62\frac{1}{2}\% = 5/8$ · $66\frac{2}{3}\% = 2/3$ · $83\frac{1}{3}\% = 5/6$ · $8\frac{1}{3}\% = 1/12$ · $14\frac{2}{7}\% = 1/7$ · **an increase of $16\frac{2}{3}\%$ means $\times 7/6$**

PERCENTAGE — THE FOUR RESULTS A is $x\%$ more than B » B is $x/(100 + x)$ less than A (25% more » 20% less) · up $x\%$ then down $x\%$ » **loss of $x^2/100$** · price up $x\%$ » cut consumption by $x/(100 + x)$ · successive discounts a and b » **$a + b - ab/100$** . **PERCENTAGES DO NOT ADD — MULTIPLY THE FACTORS**

RATIO find what **one part** is worth first · a difference of shares is a difference of **parts** · combine ratios by making the common term equal · when adding to a mixture, **anchor on the unchanged component**

AVERAGE total = average $\times$ number · **change in total = number $\times$ change in average** · overlapping pairs: **sum of pairs - whole = the overlap** · a misread observation shifts the average by **error $\div$ n**

ALLIGATION cheaper : dearer = (dearer - mean) : (mean - cheaper), **crosswise** · always alligate on **one component** · repeated replacement: **$x[(x - y)/x]^n$ remains**

AGES use one variable **$7x$ and $3x$** · add the shift to **every** age · **substitute back to check** · ratios move **towards 1 : 1** with time · the trap is x , or the wrong person's age

TIME AND WORK total work = LCM of the days · add **rates**, never days · twice as efficient = half the time · **wages divide in the ratio of work DONE** · emptying pipe = negative rate · $M_1D_1 = M_2D_2$

SPEED average speed = total distance $\div$ total time · equal **distances** » harmonic mean **$2uv/(u + v)$** · equal **times** » arithmetic mean · km/h to m/s $\times 5/18$

PROFIT always on **cost** · discount on **marked price** · same SP with $+x\%$ and $-x\%$ » **always a loss of $x^2/100$**

INTEREST SI = $PRT/100$ · CI difference over **2 years = $P(R/100)^2$** · half-yearly: rate halves, time doubles · SI doubles money when $RT = 100$

11. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F8.5

$16\frac{2}{3}\% = 1/6 \cdot 12\frac{1}{2}\% = 1/8 \cdot 37\frac{1}{2}\% = 3/8 \cdot 66\frac{2}{3}\% = 2/3$ — CONVERT BEFORE YOU MULTIPLY

25% MORE ONE WAY IS 20% LESS THE OTHER. THE BASE MOVES

UP 20% THEN DOWN 20% = DOWN 4%, NEVER ZERO

PRICE UP 25% » CONSUMPTION DOWN 20%

DISCOUNTS 20% AND 10% = 28%, NOT 30%

FIND WHAT ONE PART IS WORTH

A DIFFERENCE BETWEEN SHARES IS A DIFFERENCE BETWEEN PARTS

TOTAL = AVERAGE × NUMBER. CHANGE IN TOTAL = $n \times$ CHANGE IN AVERAGE

PAIRS MINUS WHOLE = THE OVERLAP

ALLIGATE CROSSWISE, ON ONE COMPONENT ONLY

TAKE THE LCM AS THE TOTAL WORK — THEN NO FRACTIONS APPEAR

ADD RATES, NEVER DAYS. WAGES FOLLOW WORK DONE

AGES: ONE VARIABLE, SHIFT EVERY AGE, SUBSTITUTE BACK

EQUAL DISTANCES » $2uv/(u + v)$. 40 AND 60 GIVES 48, NOT 50

PROFIT IS ON COST. SAME PRICE, +x% AND -x% » ALWAYS A LOSS

CI - SI OVER TWO YEARS = $P(R/100)^2$

RE-READ WHOSE AGE, WHICH SHARE, WHAT WAS ASKED

ONE LINE FOR THE INTERVIEW Almost every arithmetic error in an aptitude paper is a base error rather than a computation error — the candidate divides by the wrong quantity — which is why naming the base before dividing is worth more than speed at multiplication.

ACTION · NEXT

F8.6 — Mensuration and Geometry. It carries the most 2023 items of any Quant chapter: the **hemisphere**, the **melted disc**, **circle geometry** and **heights and distances** were all asked.

Tonight's drill, per F8.0's week 1 Tuesday cell: **§8 Q1–12, timed at twelve minutes.**